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Summary

Three decades in software, the last nine architecting and shipping identity-protection systems at scale. Strongest where high-level systems architecture meets low-level implementation — built a company's first SSO/JWT auth platform, founded its cross-team architecture function, and led a multi-year greenfield platform rebuild that now serves ~1.7M customers. Currently driving field-level encryption, audit log standards, and AI-augmented engineering practices at Allstate Identity Protection.

Experience

Principal Software Engineer · **Allstate Identity Protection**

Jul 2022 - present · Remote (Eugene, OR)

Led the post-acquisition greenfield platform rebuild across the finish line; **platform shipped Jan 2023, serving ~1.7M customers**. Continues as a lead principal in the Solution Architecture and Design (SAD) function, driving cross-team architecture and standards.

- **Field Level Encryption:** drives both the standard and the platform-wide rollout, encrypting **~60% of customer profile fields** to protect PII against service-layer leakage — addressing what database-level encryption-at-rest leaves uncovered (compliance met; live-path leaks unmitigated).
- **Audit Log standards & implementation:** owns the org-wide standard and its rollout across services for compliance and observability.
- **Unified interface pattern:** original architecture design that exposes **CQRS and REST** through a single specification (shipped, in production), with **Model Context Protocol (MCP)** as the planned third leg now in active design — positioning AI-agent integration as a first-class interface modality alongside existing consumers.
- **AI-assisted code generation** (specification-driven, test-gated): authored a methodology in which standards inform tests, tests are human-reviewed, and an LLM produces code bound to those tests — treating the LLM as a constrained implementer, not a free agent.
- Established the company's **internal data archiving system, API standards, and security standards** as part of the post-launch SAD agenda.

Senior Developer · InfoArmor / **Allstate Identity Protection**

Jun 2017 - Jul 2022 · Phoenix, AZ

- **Built the company's first SSO / JWT identity & authentication platform** from scratch — peak ~40,000 active users — establishing the auth substrate that carried the product through multiple platform generations.
- **Drove engineering-practice modernization** in the first two years — including the move from git-based deployment to a Jenkins CI/CD pipeline — bringing standard practices into a small-company org, slowly and by partnership rather than mandate.
- **Led the post-acquisition greenfield platform rebuild** (2018+): new CI on **Buildkite** in place of Jenkins, **Helm-chart deployment to Kubernetes**, and **migration of the in-house auth platform onto Auth0** (freeing engineering capacity from owning a security-critical undifferentiated service).
- **Modernized internal API systems** as a multi-year thread — pattern work that later became the org-wide API standards under SAD.
- **Founded the Solution Architecture and Design (SAD) function:** started as an internal initiative ~late 2019; recognized as an official function within months. **Every initiative now flows through**

SAD for spec research and acceptance criteria; the function also maintains architectural alignment and interoperability across domains and teams, so engineering work is *informed*, not *ad hoc*.

Owner · Up9 Consulting

Jan 2008 - Dec 2021 · Phoenix, AZ · concurrent side practice

- Served primarily K-12 and higher-ed clients across OS X / Linux server deployment, directory and SSO integration, SIS integration and reporting, LAMP application development, MySQL data warehousing, and network-based lab imaging.
- **Remote Linux host deployments for a new PayPal datacenter** as part of a three-person team over six months.
- **AESD email migration to Google Workspace** — cleaned up an overloaded legacy mail system; included server placement and an OAuth gateway shim for legacy clients during cutover.
- **VUSD Curriculum Planner** — centralized web app that loaded curriculum from multiple districts and pushed daily/weekly goals to teachers as iCal calendar updates.
- **PVUSD LDAP centralization** — led the move of district authentication onto a single LDAP service, integrating service-by-service across the existing application landscape.
- **HCLC charter-district consolidation** — collapsed a charter district's auth and file-storage workloads onto a single high-capacity VM host; on-site deployment included.
- **PVUSD pCHAT** — Jabber server installation tied into LDAP for authentication, with MySQL-backed maintenance and persistence.

Senior Customer Delivery Technical Engineer · CyberSponse

Jun 2013 - Dec 2016 · Phoenix, AZ

Hired as a senior application developer on a Symfony2 platform; scope expanded across operations, system architecture, and lead developer duties as the startup evolved.

- **Designed the security mechanism for inter-service communications and the licensing system**, owning the cryptographic boundaries between platform components.
- **Led the application architecture rework into microservices** for geographically distributed scaling.
- Took on and delivered a major government client at end-of-year before the company closed its Phoenix office.

Earlier roles

Unix/Linux Administrator at Bechtel (May 2007 - Feb 2008) — monitoring and audit tooling on HP-UX.

Senior Systems Analyst at Arizona State University (May 2006 - May 2007) — development and project lead on a data warehouse contracted to Paradise Valley USD; aggregated data nightly from ~40 source databases (DBF, SyBase, MySQL) into a Drupal-fronted query environment with a self-service query module so end users could build views without ASU assistance.

Programmer II → Network Applications Specialist III at Paradise Valley USD (Jan 2001 - May 2006) — established the district's data warehousing initiative (later continued under ASU); built a PHP standards-based testing platform (DAPS), a 5-server OS X + XRAID web-storage cluster on Apache + WebObjects (ePAC), a district gradebook with nightly SIS sync across five high schools (pOGB), an automated account-generation system pulling from ~30 dBase sources plus Powerschool exports (pACC), a PHP trouble-ticket system, and a Moodle deployment used as a standards-testing pilot. Trained other programmers and introduced district-wide use of blogging, wikis, instant messaging, secure email, and mailing-list collaboration during a period when those were novel for a K-12 IT organization.

Junior Developer at What's For Free Technologies (May - Dec 2000) — JSP brand-promotion portal with newsletter opt-in, Oracle data-entry web frontends, and a JSP + Sendmail mailing system; the dot-com collapsed mid-December 2000.

Self-Employed Consultant (Jan 1997 - Jun 2002) — dynamic shopping carts, small-to-mid network buildouts, home-to-office VPNs.

Selected Projects

Active personal work; full notes at pknull.ai/projects.

- **Asha** — multi-harness AI workflow system: persistent identity, session memory, and pluginized workflows for Claude Code and Codex.
- **Thallus** — local-first AI agent infrastructure umbrella covering Egregore, Familiar, Servitor, Scry, and thallus-core.

Skills

- **Languages** — Python (3.10+), Go, Rust, TypeScript, JavaScript (Node.js), PHP, Bash, SQL
- **AI / LLM engineering** — Anthropic Claude API, prompt engineering, LLM-bound TDD methodology, multi-agent / multi-harness coordination, Model Context Protocol (MCP) design
- **Architecture** — CQRS, REST, event-driven systems, microservices, identity & SSO design, schema- and standards-driven design
- **Platform & infra** — Linux, Kubernetes, Helm, Docker, Apache, Buildkite, Jenkins, Ansible, AWS, Auth0, PostgreSQL, MySQL, MongoDB, Redis
- **Security** — field-level encryption, SSO / JWT, LDAP, identity & access management, audit log standards, security architecture
- **Practices** — test-driven development, cross-team standards authorship, technical writing, code review